

UNIVERSITY OF JAMMU

BHADERWAH CAMPUS



BACHELORS OF COMPUTER APPLICATIONS

(BCA-4th SEMESTER 2024-28)

PRACTICAL FILE

PYTHON PROGRAMMING

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Submitted To:

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I would like to express my gratitude to all those people who helped me to complete this practical file. I want to thank **Department of computer science and IT** for providing me all the tools and resources for the completion of my practical file.

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THANK YOU

Sajad Ahmed

BCA 4th semester

413040011

CERTIFICATE

This is to certify that **Sajad Ahmed** student of **BCA 4th semester** bearing university **Roll no. 413040011** has completed that required number of practical of **python programming** under the guidance of class teacher **Mr.Abhay khajuri** from the Department of Computer science and IT, Bhardwah campus, University of Jammu.

Signature.

(I/C course coordinate)

Signature.

(Practical I/C)

Program 1: Interactive Shell Expressions

```
print(2 + 3)
```

```
print(10 / 2)
```

```
print(2 ** 8)
```

```
print(22 % 7)
```

output:

```
5
5.0
256
1
```

Program 2: Integer, Float & String Data Types

```
age = 34
```

```
gpa = 9.5
```

```
name = "Sajad"
```

```
print(type(age))
```

```
print(type(gpa))
```

```
print(type(name))
```

```
print(age, gpa, name)
```

Output:

```
<class 'int'>
<class 'float'>
<class 'str'>
34 9.5 Sajad
```

Program 3: String Concatenation & Replication

```
first = "Hello"
```

```
second = "World"
```

```
print(first + " " + second)
```

```
print("Ha" * 3)
```

```
print("-" * 30)
```

Output:

```
Hello World
HaHaHa
```

Program 4: Storing Values in Variables

```
length = 10
```

```
width = 5
```

```
area = length * width
```

```
perimeter = 2 * (length + width)
```

```
print("Area:", area)
```

```
print("Perimeter:", perimeter)
```

Output:

```
Area: 50
Perimeter: 30
```

Program 5: Your First Program – Hello World + User Input

```
print("Hello, World!")
```

```
name = input("Enter your name: ")
```

```
print("Welcome,", name + "!")
```

```
Hello, World!  
Enter your name: Sajad  
Welcome, Sajad!
```

Program 6: Dissecting a Program – Simple Calculator

```
a = float(input("Enter first number: "))  
b = float(input("Enter second number: "))
```

```
print("Addition:", a + b)  
print("Subtraction:", a - b)  
print("Multiplication:", a * b)  
if b != 0:  
    print("Division:", a / b)  
else:  
    print("Division by zero is undefined.")
```

```
Addition: 90.0  
Subtraction: 0.0  
Multiplication: 2025.0  
Division: 1.0
```

Program 7: Temperature Converter (F to C)

```
fahrenheit = float(input("Enter temperature in Fahrenheit: "))  
celsius = (fahrenheit - 32) * 5 / 9  
print(f"{fahrenheit}°F = {celsius:.2f}°C")
```

```
Enter temperature in Fahrenheit: 34  
34.0°F = 1.11°C
```

Program 8: Simple Interest Calculator

```
principal = float(input("Enter principal: "))  
rate = float(input("Enter rate of interest: "))  
time = float(input("Enter time (years): "))
```

```
si = (principal * rate * time) / 100  
print(f"Simple Interest = {si:.2f}")
```

```
Enter principal: 45.4  
Enter rate of interest: 34.5  
Enter time (years): 435.6  
Simple Interest = 6822.80
```

Program 9: Boolean Values & Comparison Operators

```
x = 10
```

```
y = 20
```

```
print(x > y) # False
```

```
print(x < y) # True
```

```
print(x == y) # False
```

```
print(x != y) # True
```

```
print(x >= 10) # True
```

Output

```
False  
True  
False  
True  
True
```

Program 10: Boolean Operators (and, or, not)

```
age = 20
```

```
has_id = True
```

```
if age >= 18 and has_id:
```

```
    print("Access granted.")
```

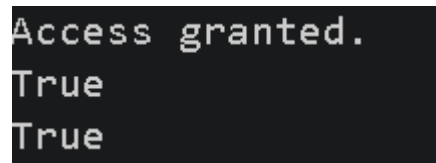
```
else:
```

```
    print("Access denied.")
```

```
print(not False)
```

```
print(True or False)
```

Output:



```
Access granted.  
True  
True
```

Program 11: Flow Control – if / elif / else

```
marks = int(input("Enter marks: "))
```

```
if marks >= 90:
```

```
    print("Grade: A")
```

```
elif marks >= 75:
```

```
    print("Grade: B")
```

```
elif marks >= 60:
```

```
    print("Grade: C")
```

```
elif marks >= 40:
```

```
    print("Grade: D")
```

```
else:
```

```
    print("Grade: F")
```



```
Grade: D
PS C:\Users\shame\OneDrive\Documents\Sajad.python> 67
67
PS C:\Users\shame\OneDrive\Documents\Sajad.python> 45
45
```

Program 12: while Loop – Countdown

```
count = 10
while count >= 0:
    print(count)
    count -= 1
print("Blast off!")
```

```
3
2
1
0
Blast off!
```

Program 13: for Loop – Multiplication Table

```
num = int(input("Enter a number: "))
for i in range(1, 11):
    print(f"{num} x {i} = {num * i}")
```

```
45 x 6 = 270
45 x 7 = 315
45 x 8 = 360
45 x 9 = 405
45 x 10 = 450
```

Program 14: Importing Modules

```
import math
import random
```

```
print("Square root of 144:", math.sqrt(144))  
print("Value of PI:", math.pi)  
print("Random number (1-100):", random.randint(1, 100))
```

Output:

```
Square root of 144: 12.0  
Value of PI: 3.141592653589793  
Random number (1-100): 2
```

Program 15: Ending a Program with sys.exit()

```
import sys
```

```
password = input("Enter password: ")
```

```
if password != "secret123":
```

```
    print("Wrong password! Exiting.")
```

```
    sys.exit()
```

```
print("Welcome to the system!")
```

Output:

```
Enter password: secret123  
Welcome to the system!
```

Program 16: Functions – def, Parameters & Return Values

```
def greet(name):
```

```
    return f"Hello, {name}! How are you?"
```

```
def add(a, b):
```

```
    return a + b
```

```
print(greet("Alice"))
```

```
print("Sum:", add(5, 3))
```

Output:

```
Hello, Sajad! How are you?  
Sum: 8
```

Program 17: Local & Global Scope + global Statement

```
count = 0
```

```
def increment():
```

```
    global count
```

```
    count += 1
```

```
    print("Inside function, count =", count)
```

```
increment()
```

```
increment()
```

```
print("Outside function, count =", count)
```

Output:

```
Inside function, count = 1  
Inside function, count = 2  
Outside function, count = 2
```

Program 18: Exception Handling

```
def safe_divide(a, b):
```

```
    try:
```

```
        result = a / b
```

```
        return result
```

```
    except ZeroDivisionError:
```

```
        return "Error: Cannot divide by zero!"
```

```
    except TypeError:
```

```
return "Error: Please enter numbers only!"
```

```
print(safe_divide(10, 2))
```

```
print(safe_divide(10, 0))
```

```
print(safe_divide(10, "a"))
```

OutPut:

```
5.0
Error: Cannot divide by zero!
Error: Please enter numbers only!
```

c Program 19: List Data Type – Creation & Access

```
fruits = ["apple", "banana", "cherry", "date", "elderberry"]
```

```
print("First fruit:", fruits[0])
```

```
print("Last fruit:", fruits[-1])
```

```
print("Fruits 2 to 4:", fruits[1:4])
```

```
print("Total fruits:", len(fruits))
```

```
First fruit: apple
Last fruit: elderberry
Fruits 2 to 4: ['banana', 'cherry', 'date']
Total fruits: 5
```

Program 20: Working with Lists – Add, Remove, Sort

```
numbers = [5, 3, 8, 1, 9, 2]
```

```
numbers.append(7)
```

```
numbers.insert(0, 0)
```

```
numbers.remove(3)
```

```
numbers.sort()
```

```
print("Sorted list:", numbers)
print("Max:", max(numbers))
print("Min:", min(numbers))
print("Sum:", sum(numbers))
```

Output:

```
Sorted list: [0, 1, 2, 5, 7, 8, 9]
Max: 9
Min: 0
Sum: 32
```

Program 21: Augmented Assignment Operators

```
x = 10
x += 5; print("+=", x)
x -= 3; print("-=", x)
x *= 2; print("*=", x)
x //= 4; print("//=", x)
x **= 2; print("**=", x)
x %= 5; print("%=", x)
```

```
-= 12
*= 24
//= 6
**= 36
%= 1
```

Program 22: List Methods – Comprehensive Demo

```
items = [3, 1, 4, 1, 5, 9, 2, 6, 5]

print("Count of 1:", items.count(1))
print("Index of 9:", items.index(9))
```

```
items.reverse()

print("Reversed:", items)


items.sort()

print("Sorted:", items)


popped = items.pop()

print("Popped:", popped, "| List:", items)


copy = items.copy()

print("Copy:", copy)


items.clear()

print("After clear:", items)
```

```
Reversed: [5, 6, 2, 9, 5, 1, 4, 1, 3]
Sorted: [1, 1, 2, 3, 4, 5, 5, 6, 9]
Popped: 9 | List: [1, 1, 2, 3, 4, 5, 5, 6]
Copy: [1, 1, 2, 3, 4, 5, 5, 6]
After clear: []
```

Program 23: List Comprehension – Even Numbers & Squares

```
evens = [x for x in range(1, 21) if x % 2 == 0]

print("Evens:", evens)


squares = [x**2 for x in range(1, 11)]

print("Squares:", squares)


words = ["hello", "world", "python"]

upper_words = [w.upper() for w in words]

print("Uppercase:", upper_words)
```

```
Evens: [2, 4, 6, 8, 10, 12, 14, 16, 18, 20]
Squares: [1, 4, 9, 16, 25, 36, 49, 64, 81, 100]
Uppercase: ['HELLO', 'WORLD', 'PYTHON']
```

Program 24: Dictionary Data Type – Create & Access

```
student = {
    "name": "sajad",
    "age": 20,
    "grade": "A",
    "subject": "Python"
}

print("Name:", student["name"])
print("Age:", student.get("age"))
print("Keys:", list(student.keys()))
print("Values:", list(student.values()))
```

```
Name: sajad
Age: 20
Keys: ['name', 'age', 'grade', 'subject']
Values: ['sajad', 20, 'A', 'Python']
```

Program 25: Dictionary Operations – Add, Update, Delete

```
inventory = {"apples": 10, "bananas": 5, "oranges": 8}

inventory["grapes"] = 12
inventory["apples"] = 15
del inventory["bananas"]

print("Updated inventory:", inventory)
print("Has oranges?", "oranges" in inventory)
```

```
Updated inventory: {'apples': 15, 'oranges': 8, 'grapes': 12}
Has oranges? True
```

Program 26: Pretty Printing with pprint

```
import pprint
```

```
data = {
    "students": [
        {"name": "Alice", "marks": 95},
        {"name": "Bob", "marks": 88},
        {"name": "Charlie", "marks": 76}
    ],
    "class": "Python 101",
    "teacher": "Dr. Smith"
}
```

```
pprint.pprint(data)
```

```
{'class': 'Python 101',
 'students': [{'marks': 95, 'name': 'Alice'},
               {'marks': 88, 'name': 'Bob'},
               {'marks': 76, 'name': 'Charlie'}],
 'teacher': 'Dr. Smith'}
```

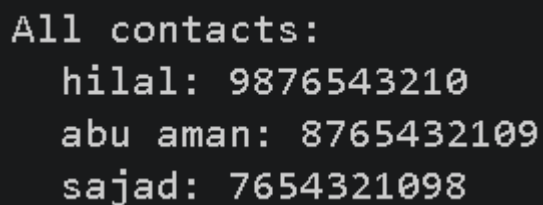
Program 27: Using Dictionaries to Model Real-World Things

```
phonebook = {}
```

```
def add_contact(name, phone):
    phonebook[name] = phone
```



```
def lookup(name):  
    return phonebook.get(name, "Contact not found.")  
  
add_contact("hilal", "9876543210")  
add_contact("abu aman", "8765432109")  
add_contact("sajad", "7654321098")  
  
print("sajad's number:", lookup("sajad"))  
print("shadab's number:", lookup("shadab"))  
  
print("\nAll contacts:")  
for name, phone in phonebook.items():  
    print(f" {name}: {phone}")
```

A terminal window with a dark background and light green text. It displays the output of the Python script, showing all contacts in a dictionary-like format.

```
All contacts:  
hilal: 9876543210  
abu aman: 8765432109  
sajad: 7654321098
```

Program 28: Working with Strings – Slicing & Indexing

```
text = "Hello, Python Programming!"
```

```
print("Length:", len(text))  
print("Uppercase:", text.upper())  
print("Lowercase:", text.lower())  
print("Slice [7:13]:", text[7:13])  
print("Reversed:", text[::-1])  
print("Replace:", text.replace("Python", "World"))
```

```
Uppercase: HELLO, PYTHON PROGRAMMING!  
Lowercase: hello, python programming!  
Slice [7:13]: Python  
Reversed: !gnimmargorP nohtyP ,olleH  
Replace: Hello, World Programming!
```

Program 29: Useful String Methods

```
sentence = " Welcome to Python Programming! "
```

```
print(sentence.strip())  
print(sentence.strip().split())  
print("python" in sentence.lower())  
print(sentence.strip().startswith("Welcome"))  
print(sentence.strip().endswith("!"))  
print(sentence.strip().count("o"))  
print(",".join(["a", "b", "c"]))  
print("hello".capitalize())  
print("hello world".title())  
print("abc123".isalnum())
```

```
4  
a,b,c  
Hello  
Hello World  
True
```

Program 30: Capstone Program – Student Report Card (Uses ALL 4 Units)

```
import pprint  
  
def get_grade(marks):  
    if marks >= 90: return "A"  
    elif marks >= 75: return "B"
```

```
elif marks >= 60: return "C"
```

```
elif marks >= 40: return "D"
```

```
else: return "F"
```

```
def generate_report(name, marks_list):
```

```
    subjects = ["Math", "Science", "English", "Python", "History"]
```

```
    report = {
```

```
        "student": name,
```

```
        "marks": {},
```

```
        "grades": {}
```

```
    }
```

```
    for i, subject in enumerate(subjects):
```

```
        report["marks"][subject] = marks_list[i]
```

```
        report["grades"][subject] = get_grade(marks_list[i])
```

```
    total = sum(marks_list)
```

```
    avg = total / len(marks_list)
```

```
    report["total"] = total
```

```
    report["average"] = round(avg, 2)
```

```
    report["overall_grade"] = get_grade(avg)
```

```
    return report
```

```
name = input("Enter student name: ")
```

```
marks = []
```

```
for sub in ["Math", "Science", "English", "Python", "History"]:
```

```
    m = int(input(f"Enter marks for {sub}: "))
```

```
    marks.append(m)
```

```
Enter student name: sajd
Enter marks for Math: 567
Enter marks for Science: 54
Enter marks for English: 5
Enter marks for Python: 6567
Enter marks for History: 67
```

```
==== REPORT CARD ====
{'average': 1452.0,
 'grades': {'English': 'F',
            'History': 'C',
            'Math': 'A',
            'Python': 'A',
            'Science': 'D'}},
'marks': {'English': 5,
          'History': 67,
          'Math': 567,
          'Python': 6567,
          'Science': 54},
'overall_grade': 'A',
'student': 'sajd',
'total': 7260}
```

```
report = generate_report(name, marks)
```

```
print("\n==== REPORT CARD =====")
```

```
pprint.pprint(report)
```